Hector Lugo III

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Education

The University of Texas Rio Grande Valley Expected Dec. 2026
M.S. in Computer Science (GPA: 3.79/4.00) | B.S. in Computer Science (Aug. 2020 – May 2024)

Experience

AI/ML Research Intern May 2026 – Aug. 2026
Air Force Research Laboratory (AFRL)

- Built and open-sourced an evaluation harness for multi-camera video QA: 650K+ scored answers across 3 benchmarks and 2 vision-language models (Qwen2.5-VL, InternVL3) on a multi-GPU SLURM cluster. [code]
- Benchmarked centralized fusion vs. per-stream agents and agentic frame/clip selection (SigLIP, ViCLIP) vs. uniform sampling at matched budgets, with blind and random-control baselines to separate visual signal from answer-prior effects; manuscript in preparation.

Research Assistant Jan. 2023 – Aug. 2026
Machine Intelligence Lab, UTRGV

- Researched hierarchical control for legged robots, with a VLM navigator commanding RL locomotion policies (Unitree Go2, G1 humanoid); co-authored an accepted IEEE ICMLA 2026 paper on VLM-guided quadruped navigation.
- Built Isaac Sim and MuJoCo environments and trained PPO locomotion and navigation policies for the Unitree Go2, including state autoencoders that cut compute cost and sped up training.

Machine Learning Engineer Intern May 2025 – June 2026
Idaho National Laboratory (INL)

- Engineered a ResNet autoencoder to denoise smart-grid sensor signals for anomaly detection.
- Cut model training time by over 93% (30 days to 1–2 days) through signal preprocessing and parallel computing.

Teaching Assistant Jan. 2024 – Present
The University of Texas Rio Grande Valley (UTRGV)

- Teach CS 1101 (Intro to CS) and Digital Image Processing to 40+ students, running lectures, labs, and grading.

Data Science Research Intern June 2023 – Aug. 2023
University of California, Riverside

- Applied spectral clustering to USDOT railway crossing accident records; published at ASME JRC 2024.

Projects

VLM-Guided Legged Robot Navigation with Revisable Memory | *VLM + RL, MuJoCo* | [code] 2026

- Frozen Qwen3-VL navigator steers a PPO locomotion policy (Go2/G1, MuJoCo); designed a revisable belief memory for out-of-view target search that drops stale beliefs on contradiction, with memoryless and frame-history ablations.

Poker RL: 6-Max No-Limit Hold'em Self-Play Agent | *PPO, Gymnasium, FastAPI* | [code] 2026

- MaskablePPO self-play agent trained against a pool of frozen policy snapshots in RLCard 6-max NLHE, evaluated in mbb/hand with 95% CIs, and served as a real-time browser poker table over FastAPI + WebSockets; CI and tests.

FlowCount: Real-Time Traffic Analytics | *YOLO11, ByteTrack, Docker* | [code] 2025

- YOLO11 detections feed a ByteTrack-style tracker (Kalman filter + Hungarian matching) for line counts, wrong-way alerts, zone occupancy, and heatmaps from webcam, video, or RTSP input; web dashboard, Docker deploy, CI.

Publications

- **IEEE ICMLA 2026.** Vision–Language Guided Quadruped Navigation with RL Control (accepted).
- **ICICT 2025.** Personalized Chemotherapy Dosing Through Offline Reinforcement Learning (first author). [paper]
- **ICICT 2025.** Offline Reinforcement Learning Approaches for Safe and Effective Smart Grid Control. [paper]
- **ASME JRC 2024.** Spectral Clustering in Railway Crossing Accidents Analysis. [paper] [code]

Technical Skills

Languages: Python, C++, Java, SQL
Frameworks & Tools: PyTorch, JAX, Hugging Face, Stable-Baselines3, OpenCV, FastAPI, Docker, Linux, SLURM
Robotics & Edge: Isaac Lab, Isaac Sim, MuJoCo, LocoMuJoCo, ROS 2, Unitree Go2, NVIDIA Jetson Nano
Concepts: Reinforcement Learning, VLMs/VLAs, Multi-Agent Systems, Multi-Camera Perception, Computer Vision